

ASSUMPTIONS, LIMITATIONS, RISKS AND DEPENDENCIES

ASSUMPTIONS

The analysis will make the following assumptions:

1. Appointment_id uniquely identifies an appointment.
2. Appointment_outcome represents the final status of the appointment.
3. Previous_no_shows represents the patient's historical missed appointments.
4. Booking_lead_days represents the period between booking and the scheduled appointment.
5. A None value in reminder_channel can represent an appointment for which no reminder was sent.
6. The HealthConnect dataset is a fictional and anonymized project dataset.
7. Cancelled appointments will be treated separately from no-shows because the clinic provides cancellation and rescheduling procedures.

LIMITATIONS

- The dataset contains limited missing information in distance and waiting-time fields.
- The available variables may not capture every factor that influences whether a patient attends an appointment.
- The dataset is fictional and anonymized, so findings should be interpreted within the Experience Lab context rather than assumed to represent all real healthcare providers.
- The analysis can identify associations and patterns but cannot automatically establish causation.
- Appointment outcomes are limited to Attended, No-Show and Cancelled, so other possible patient behaviours may not be represented.

RISKS

Risk	Potential Impact	Mitigation
Duplicate appointment records	May distort counts and KPIs	Validate appointment IDs
Missing data	May affect analysis of specific variables	Assess and document missing values

Risk	Potential Impact	Mitigation
Incorrect outcome classification	May produce an inaccurate no-show rate	Keep cancellations separate
Inconsistent dates	May affect time-based analysis	Validate derived date fields
Reminder inconsistencies	May affect reminder analysis	Cross-check reminder fields
Correlation interpreted as causation	Could lead to incorrect recommendations	Clearly distinguish association from causation
Business-rule inconsistencies	Could affect findings	Validate against the Clinic Knowledge Base

DEPENDENCIES

This Project depends on:

- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- HealthConnect Clinic Knowledge Base
- Appropriate analytical tools like Power BI
- Clarification of unclear business rules where necessary

The project specifically identifies the Appointment Dataset and Data Dictionary as primary resources for Data Analytics, while the Clinic Knowledge Base provides business context such as appointment procedures and cancellation/rescheduling rules.

ETHICAL AND SAFETY CONSIDERATIONS

The dataset is anonymized, but patient-related information should still be handled responsibly. Analysis should focus on identifying patterns rather than labelling or blaming individual patients. Any future recommendations should support better appointment management and patient engagement rather than unfairly penalizing particular demographic groups.